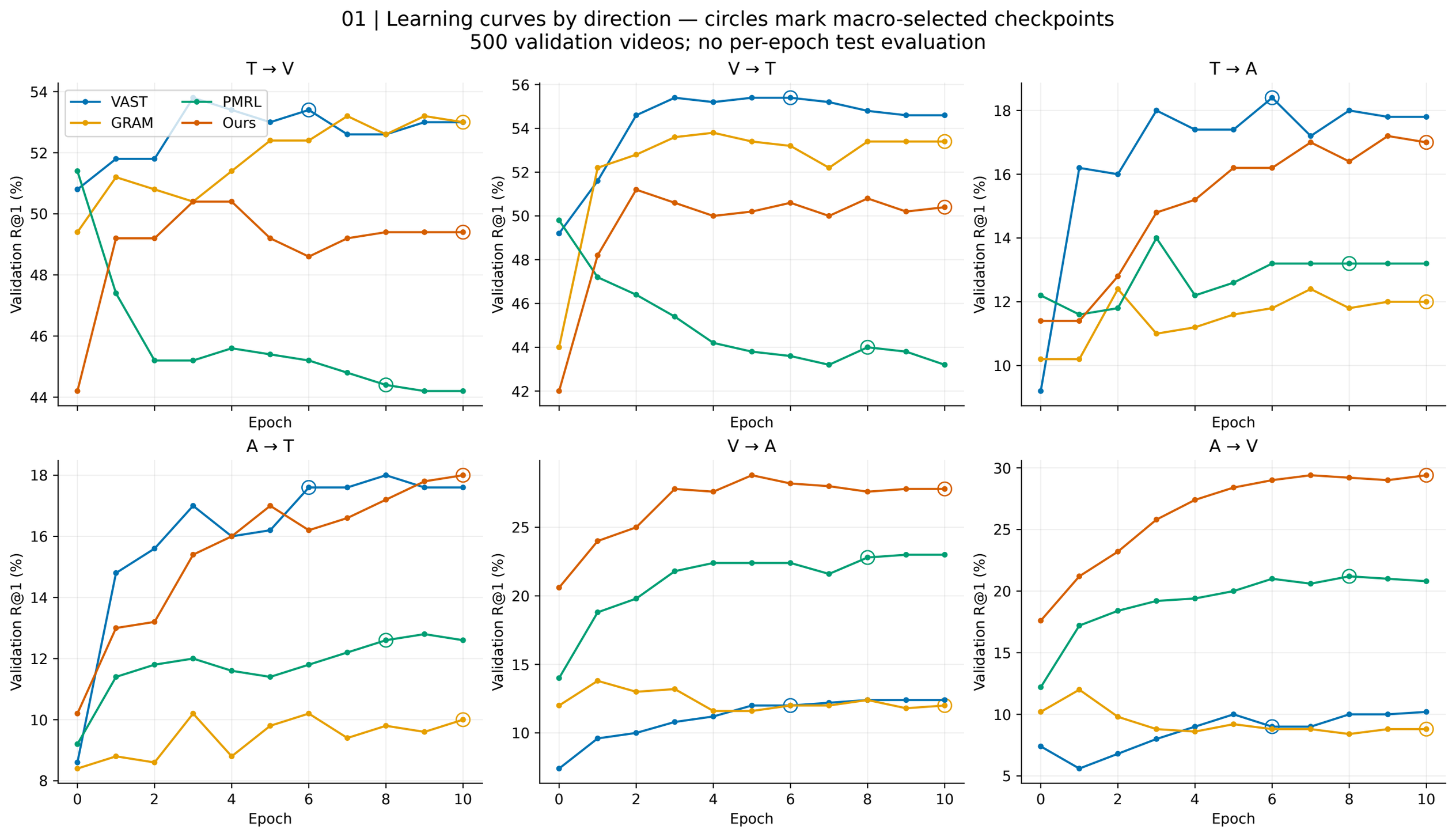
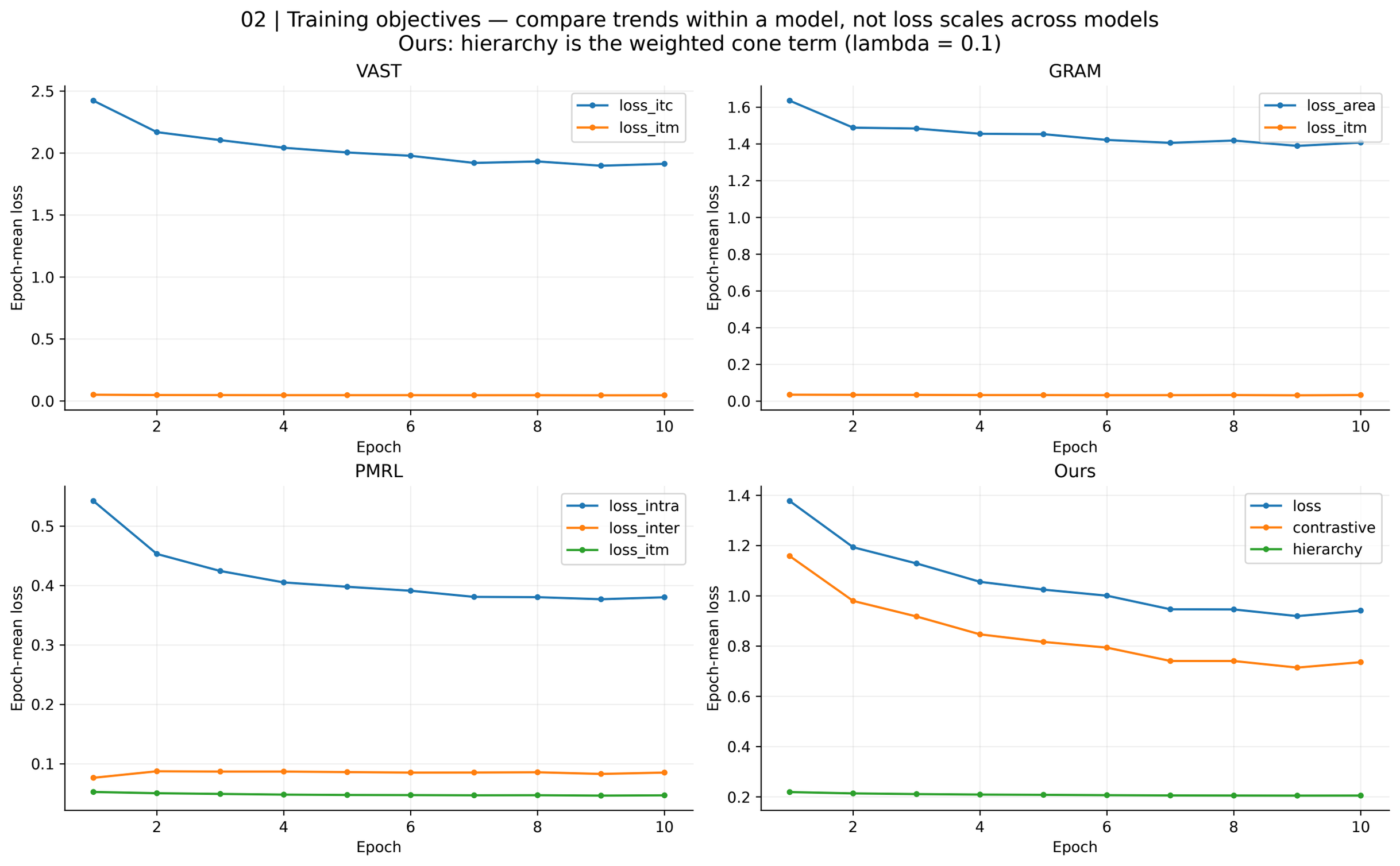
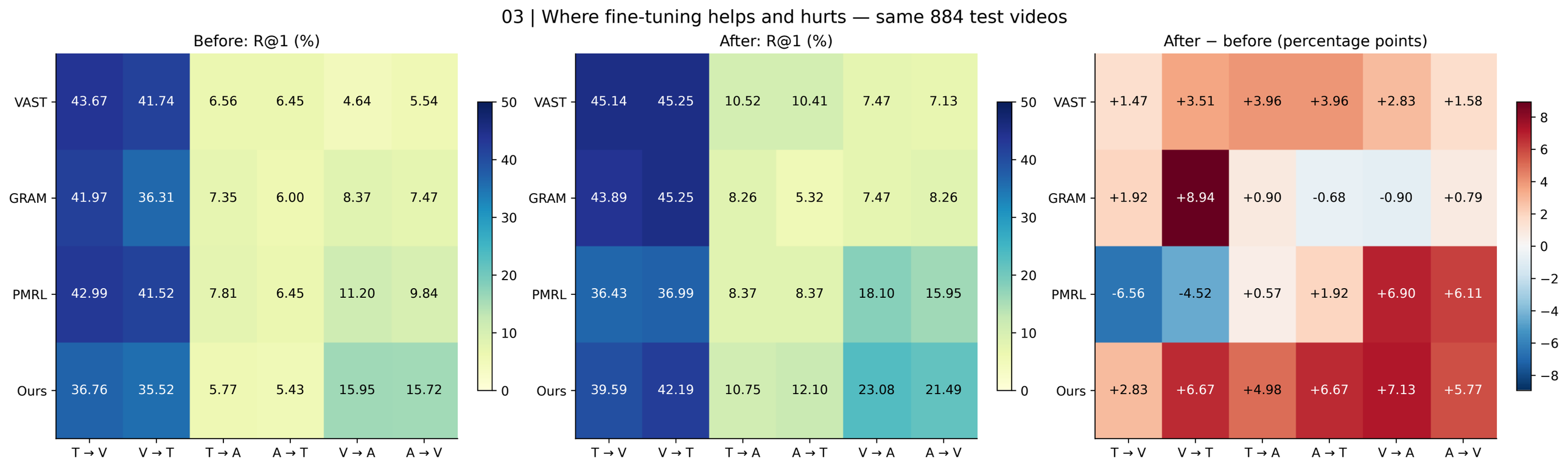
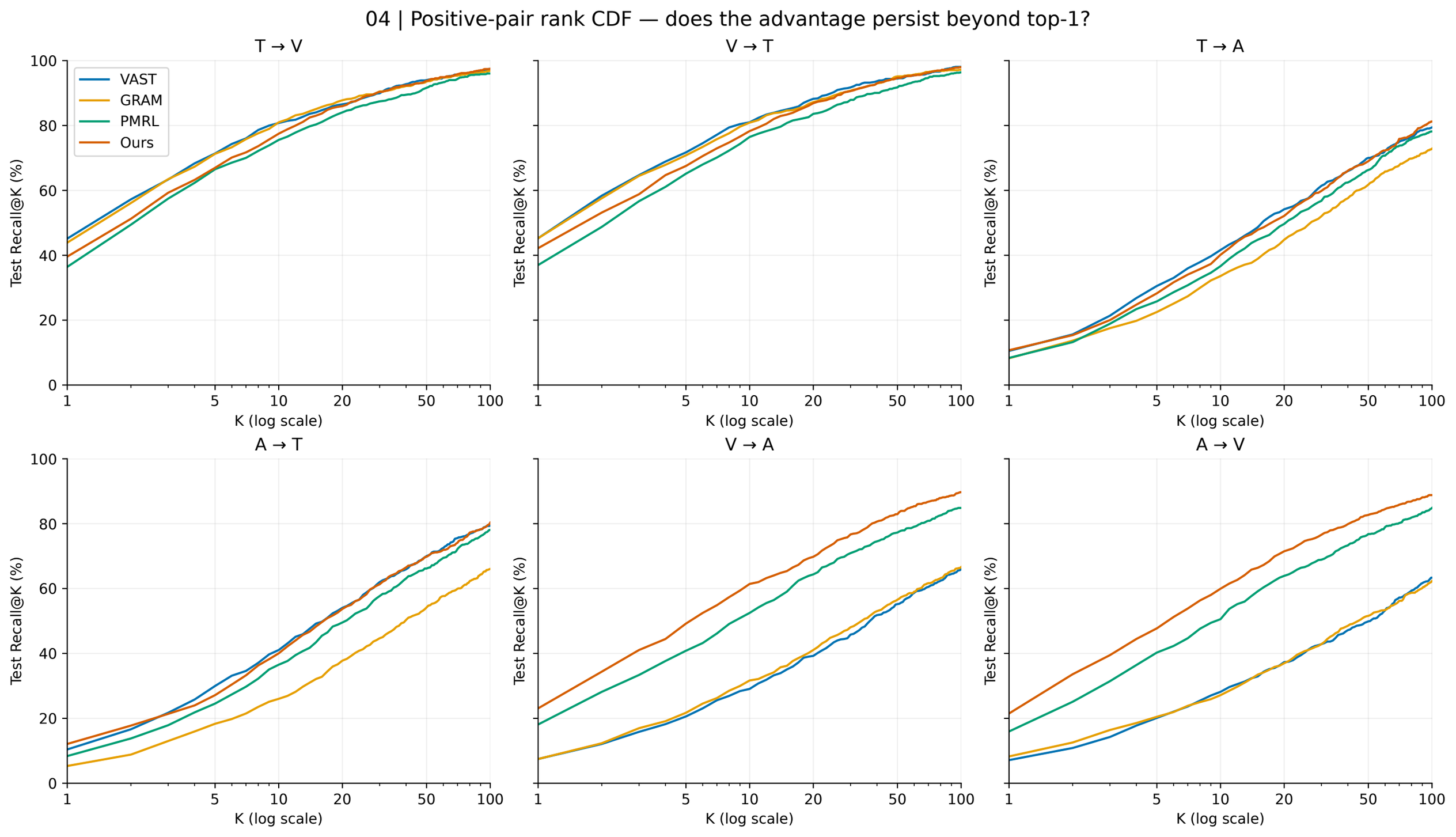
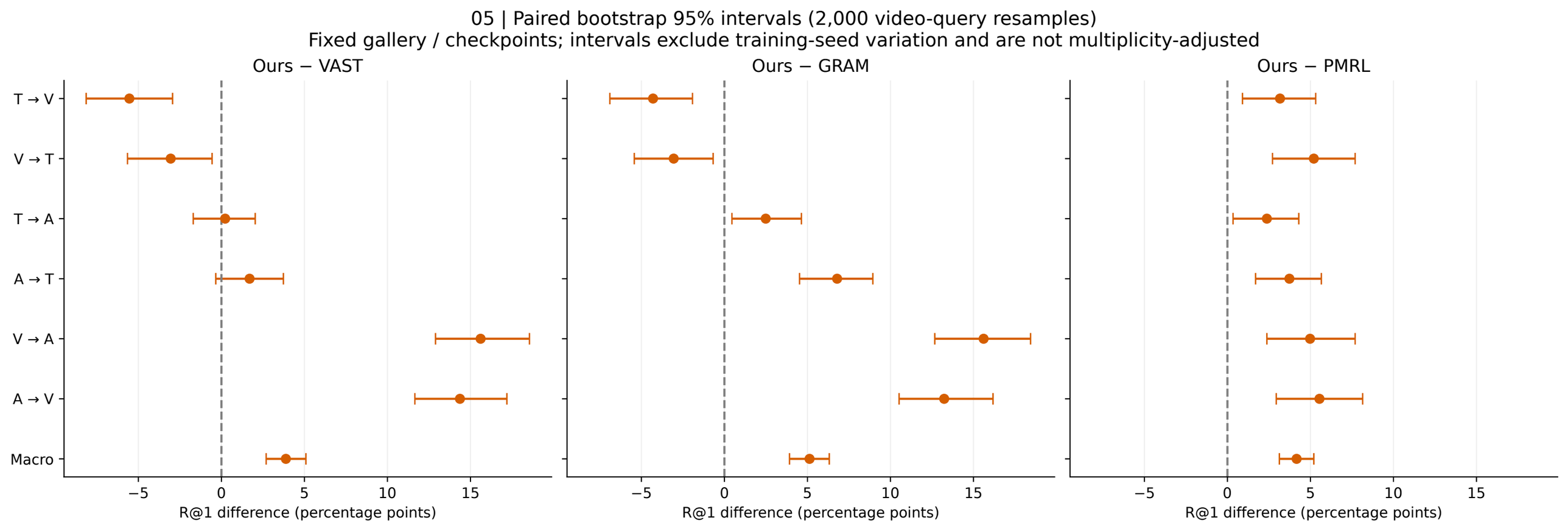
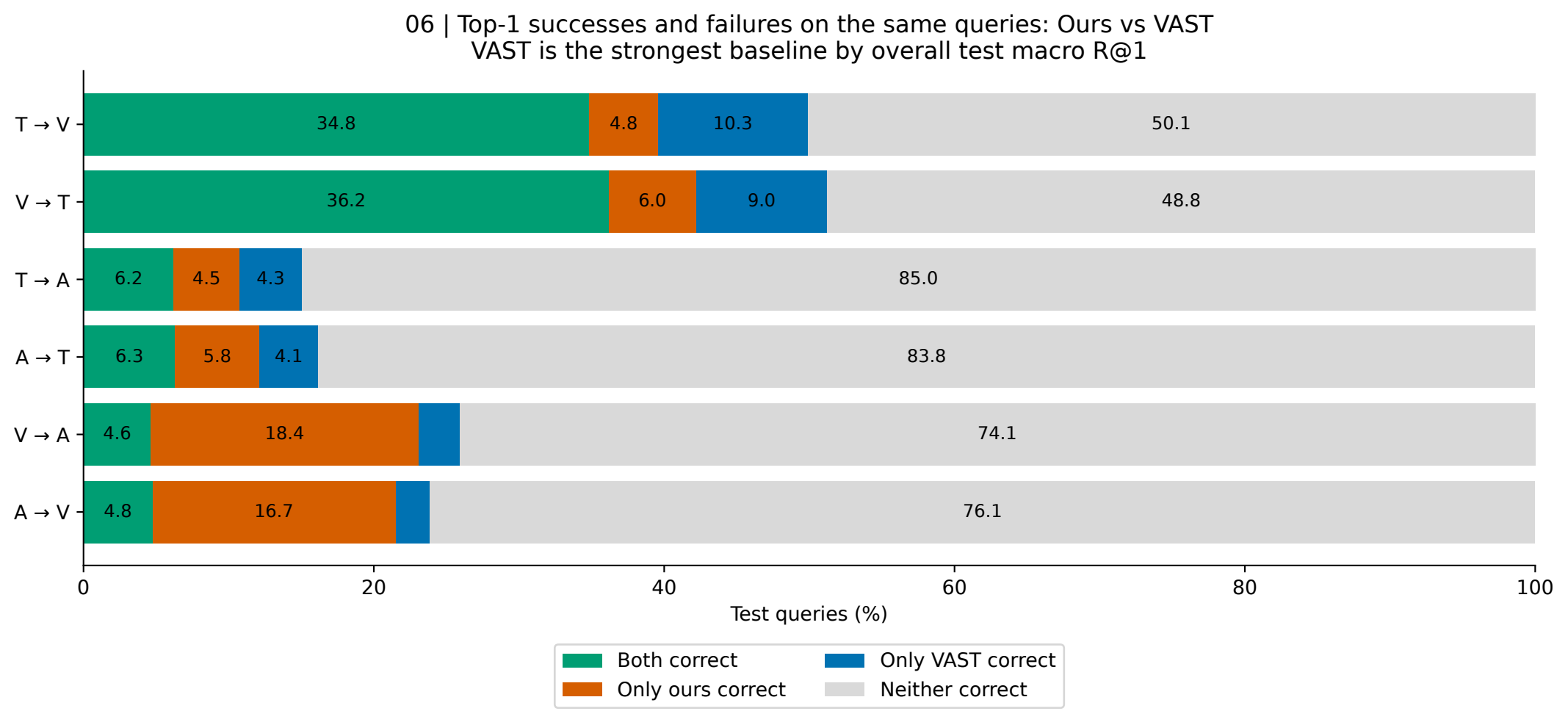

01 | Learning curves by direction — circles mark macro-selected checkpoints
500 validation videos; no per-epoch test evaluation
T → V
V → T
T → A
A → T
V → A
A → V
VAST
GRAM
PMRL
Ours
Validation R@1 (%)
Epoch

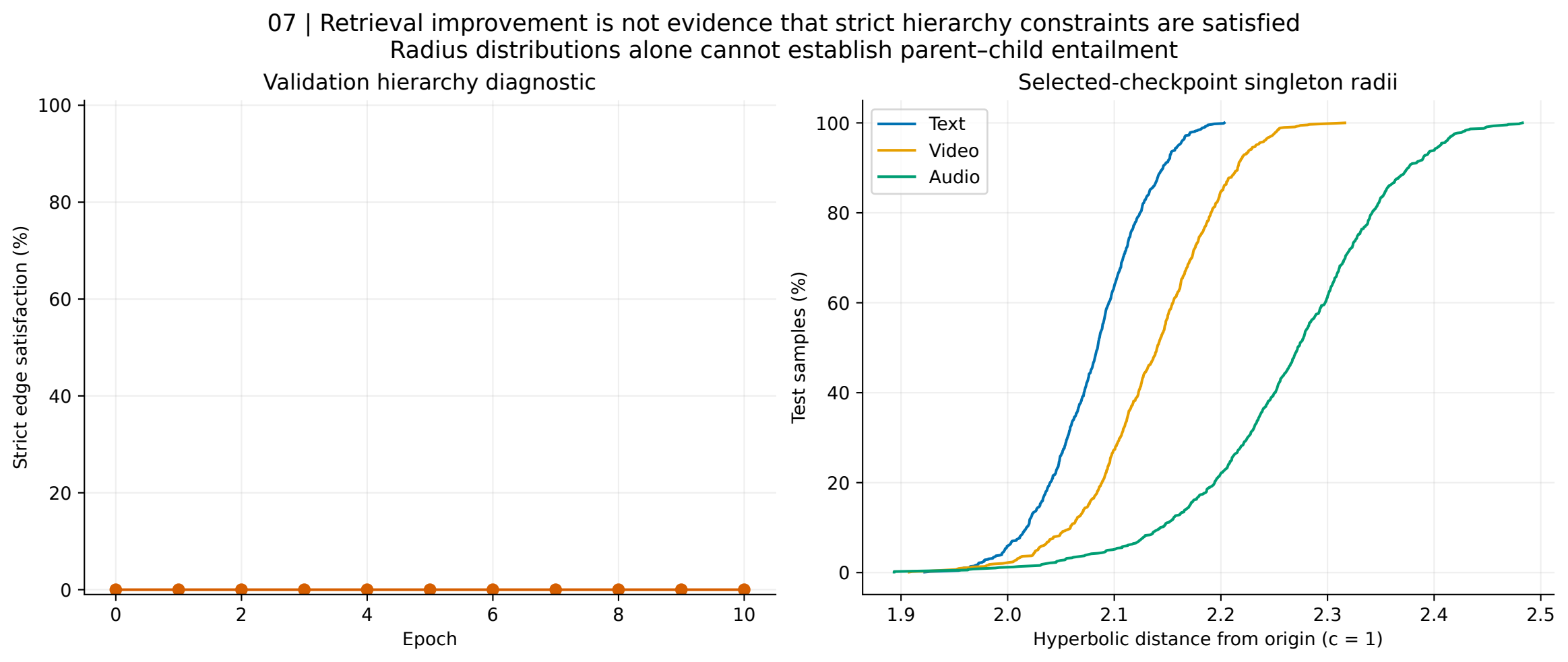

02 | Training objectives — compare trends within a model, not loss scales across models
Ours: hierarchy is the weighted cone term (lambda = 0.1)
VAST
loss_itc
loss_itm
Epoch-mean loss
Epoch
GRAM
loss_area
loss_itm
Epoch-mean loss
Epoch
PMRL
loss_intra
loss_inter
loss_itm
Epoch-mean loss
Epoch
Ours
loss
contrastive
hierarchy
Epoch-mean loss
Epoch


03 | Where fine-tuning helps and hurts — same 884 test videos
Before: R@1 (%)
After: R@1 (%)
After − before (percentage points)
VAST
GRAM
PMRL
Ours
T → V
V → T
T → A
A → T
V → A
A → V
43.67 41.74 6.56 6.45 4.64 5.54
41.97 36.31 7.35 6.00 8.37 7.47
42.99 41.52 7.81 6.45 11.20 9.84
36.76 35.52 5.77 5.43 15.95 15.72
45.14 45.25 10.52 10.41 7.47 7.13
43.89 45.25 8.26 5.32 7.47 8.26
36.43 36.99 8.37 8.37 18.10 15.95
39.59 42.19 10.75 12.10 23.08 21.49
+1.47 +3.51 +3.96 +3.96 +2.83 +1.58
+1.92 +8.94 +0.90 -0.68 -0.90 +0.79
-6.56 -4.52 +0.57 +1.92 +6.90 +6.11
+2.83 +6.67 +4.98 +6.67 +7.13 +5.77
0 10 20 30 40 50
−8 −6 −4 −2 0 2 4 6 8


04 | Positive-pair rank CDF — does the advantage persist beyond top-1?
T → V
V → T
T → A
A → T
V → A
A → V
VAST
GRAM
PMRL
Ours
Test Recall@K (%)
K (log scale)
0
20
40
60
80
100
1
5
10
20
50
100


05 | Paired bootstrap 95% intervals (2,000 video-query resamples)
Fixed gallery / checkpoints; intervals exclude training-seed variation and are not multiplicity-adjusted
Ours − VAST
Ours − GRAM
Ours − PMRL
T → V
V → T
T → A
A → T
V → A
A → V
Macro
−5
0
5
10
15
R@1 difference (percentage points)


06 | Top-1 successes and failures on the same queries: Ours vs VAST
VAST is the strongest baseline by overall test macro R@1
T → V
V → T
T → A
A → T
V → A
A → V
34.8
4.8
10.3
50.1
36.2
6.0
9.0
48.8
6.2
4.5
4.3
85.0
6.3
5.8
4.1
83.8
4.6
18.4
74.1
4.8
16.7
76.1
0
20
40
60
80
100
Test queries (%)
Both correct
Only ours correct
Only VAST correct
Neither correct


07 | Retrieval improvement is not evidence that strict hierarchy constraints are satisfied
Radius distributions alone cannot establish parent–child entailment
Validation hierarchy diagnostic
Strict edge satisfaction (%)
Epoch
Selected-checkpoint singleton radii
Test samples (%)
Hyperbolic distance from origin (c = 1)
Text
Video
Audio